

Physics of Cognition: Homework 2

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Q1

a

$$\begin{aligned} P(\text{infected} \mid \text{positive}) &= \frac{P(\text{positive} \mid \text{infected})P(\text{infected})}{P(\text{positive})} \\ &= \frac{(0.80)(0.01)}{(0.80)(0.01) + (0.05)(0.99)} \approx 0.139 \end{aligned}$$

If a certain case is positive, the probability that they are truly infected is 0.139. Sensitivity measures how well the test works on infected people to say that they have the disease while the specificity shows how well the test correctly indicates a negative result for a group that isn't actually affected (but also how much the test wrongly accuses that healthy patients have the disease).

b

$$\begin{aligned} P(\text{infected} \mid ++) &= \frac{P(\text{positive} \mid \text{infected})P(\text{infected} \mid +)}{P(+ \mid \text{infected})P(\text{infected} \mid +) + P(+ \mid \text{not infected})P(\text{not infected} \mid +)} \\ &= \frac{(0.80)(0.139)}{(0.80)(0.139) + (0.05)(1 - 0.139)} \approx 0.72 \end{aligned}$$

The previous calculation was pretty sad since we're only saying that there's a 13.9% chance that a person tested positive actually has a disease. When we apply Bayes' rule against to the tested population, we get a much better result of 72% chance that the person has the disease. However, between these calculations, the specificity and sensitivity stayed the same as that's fixed for the test specification given to us.

c

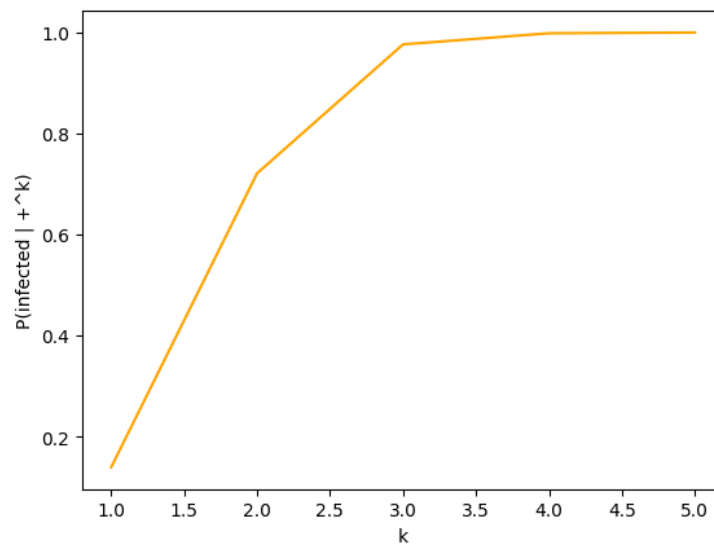
$$\begin{aligned} P(\text{infected} \mid +, -) &= \frac{P(- \mid \text{infected})P(\text{infected} \mid +)}{P(- \mid \text{infected})P(\text{infected} \mid +) + P(- \mid \text{not infected})P(\text{not infected} \mid +)} \\ &= \frac{(0.20)(0.139)}{(0.20)(0.139) + (0.95)(1 - 0.139)} \approx 0.033 \end{aligned}$$

This result shows us that even after getting a positive result, earning a negative result makes it much less likely that one is infected, so much so that it's nearing the chance that anyone from the population is truly infected. This dropped a lot from the initial 0.14 since the specificity is high, meaning it's unlikely that this person is still infected after testing negative. Yet, there's still the chance that the test missed out on an infected person as the sensitivity isn't 1.

d

```
def posterior_prob(k, s=0.8, c=0.95, p=0.01):
    f = 1 - c
    return (s**k * p) / (s**k * p + f**k * (1-p))
pos = []
for k in range(1, 6):
    pos.append(posterior_prob(k))

plt.plot(range(1, 6), pos, color="orange")
plt.xlabel("k")
plt.ylabel("P(Infected | +^k)")
```



As k increases, the posterior value sharply increases and then tapers off after $k = 3$. Changing s doesn't do much to the graph, but c can affect a lot since that leaves room for false positives and p determines the chance of the disease for the whole population.

e

1. Independence of experiments - we assume that each experiment repeatedly done on a group has no effects on one another, but this isn't true for most real tests since each test could modify how the body responds to future ones. This means that not only the priors change between each successive test on an individual but also the properties of the tests.
2. Precision of each test - we can't always ensure that the test results (even with the specificity/sensitivity in mind) are accurate as different environments, conditions, and variables might interfere with the validity which certainly could occur in the COVID-19 test sites.
3. Accuracy of the prior - in our case, the prior was constant for the total population. Yet, it's difficult to record a really accurate prior of the infection rates of COVID-19 as many infected people hadn't shown symptoms yet (which the polls would have missed).

Q2

a

$$\begin{aligned} p(v, a | x) &= p(v | x) p(a | x), \\ p(x | v, a) &\propto p(v, a | x) p(x) \\ &= p(v | x) p(a | x) p(x). \end{aligned}$$

In the brain the visual and auditory cues should have no influence on each other so they could be sampled from two different Normal distributions. The flat prior means that $p(x)$ has a proportionality of 1 with respect to the posterior so we could remove that term from the equation.

b

$$\begin{aligned} w_v + w_a &= \frac{1/\sigma_v^2}{1/\sigma_v^2 + 1/\sigma_a^2} + \frac{1/\sigma_a^2}{1/\sigma_v^2 + 1/\sigma_a^2} \\ &= \frac{1/\sigma_v^2 + 1/\sigma_a^2}{1/\sigma_v^2 + 1/\sigma_a^2} = 1 \end{aligned}$$

In this equation, the variance for each cue is inversely proportionally to its weight. With this formulation, the brain can be more confident with sources that are more reliable.

c

$$v \mid x \sim \mathcal{N}(x, \sigma^2), a \mid x \sim \mathcal{N}(x, \sigma^2)$$

$$\begin{aligned} p(v, a \mid x) &= p(v \mid x) p(a \mid x) \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(v-x)^2}{2\sigma^2}\right) \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(a-x)^2}{2\sigma^2}\right) \\ &= C \exp\left(-\frac{(v-x)^2}{2\sigma^2} - \frac{(a-x)^2}{2\sigma^2}\right) \\ \log p(v, a \mid x) &= C_2 - \frac{(v-x)^2}{2\sigma^2} - \frac{(a-x)^2}{2\sigma^2} \end{aligned}$$

Then we take the derivative and set it to 0 for max likelihood.

$$\begin{aligned} \frac{d}{dx}(\log p(v, a \mid x)) &= -\frac{d}{dx}\left(\frac{v^2 - 2vx + x^2}{2\sigma^2} + \frac{a^2 - 2ax + x^2}{2\sigma^2}\right) \\ &= -\frac{d}{dx}\left(\frac{v^2}{2\sigma^2} - \frac{vx}{\sigma^2} + \frac{x^2}{2\sigma^2} + \frac{a^2}{2\sigma^2} - \frac{ax}{\sigma^2} + \frac{x^2}{2\sigma^2}\right) \\ &= -\left(-\frac{v}{\sigma^2} + \frac{x}{\sigma^2} - \frac{a}{\sigma^2} + \frac{x}{\sigma^2}\right) = 0 \end{aligned}$$

Now let's sub in the actual sigma values and solve for x .

$$\begin{aligned} -\frac{v}{\sigma_v^2} + \frac{x}{\sigma_v^2} - \frac{a}{\sigma_a^2} + \frac{x}{\sigma_a^2} &= 0 \\ x\left(\frac{1}{\sigma_v^2} + \frac{1}{\sigma_a^2}\right) &= \frac{v}{\sigma_v^2} + \frac{a}{\sigma_a^2} \\ x &= \frac{\sigma_a^2}{\sigma_a^2 + \sigma_v^2} v + \frac{\sigma_v^2}{\sigma_a^2 + \sigma_v^2} a \end{aligned}$$

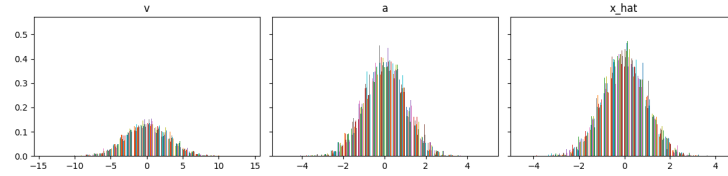
And these are our weights!

d

$$\begin{aligned} \frac{1}{1/\sigma_v^2 + 1/\sigma_a^2} &< \sigma_v^2 \\ 1 &< \sigma_v^2(1/\sigma_v^2 + 1/\sigma_a^2) = 1 + \sigma_v^2/\sigma_a^2 \\ 0 &< \sigma_v^2/\sigma_a^2 \end{aligned}$$

Since the fraction is always positive, this always holds true. By weighing the each cue by exactly the inverse of its variance, we can minimize the noise in the prediction as both distributions jointly can have less uncertainty.

e



Empirical variances:

$$\text{Var}[v] = 9.003$$

$$\text{Var}[a] = 1.005$$

$$\text{Var}[x_{\text{hat}}] = 0.904$$

Yes, the fused estimate has lower variance than each of the cues! The histogram is slightly more centered for $\hat{x}$.

f

If the prior was centered at 0, the variance of the posterior estimate would be lower, pulling $\hat{x}$ toward 0 as well.

Q3

a

In the first example, we saw how taking a test (especially if the prior is low) is not super fruitful and we should take multiple tests and repeatedly apply Bayes' rule on the select population to increase our posterior. But throughout the process, the likelihood of a positive result given an infection stayed the same. Another example of this method is stock market prediction where the posterior could be framed as the probability of increase/decrease in tomorrow's market given the state of today's. In a similar fashion to the homework problem, we can improve our predictions by incorporating multiple days into our prior—though, it's hard to quantify the dependence of the market prices between each day.

With the cue-combination, we're still assuming that by taking the maximum likelihood of both independent variables together, we can formulate the problem based on the weighted average based on the individual variances. The prior let's us center the mean ($\hat{x}$ further, yet it's unclear how to determine a distribution for the mean without more information about how the cues are processed in the brain. A real example is control systems where there are multiple sources of input that need to be processed to take an action. Fundamentally, these systems have the same Bayesian core where the optimal variable depends on variance.

b

I thought this homework was easier than the previous especially because I connected the concepts of Bayes' theorem with the activities we did in class + the

journal club and there weren't too many derivations. It took approx. 3-4 hours. I'm starting to understand why the posterior is shaped the way it is and how the theorem could be applied to different problem types. I'm still quite lacking in the statistical front since the terminology doesn't come as natively as something like Linear Algebra or Calculus.